**Statement of Purpose**

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[University name]

**Research interests.** I am applying to pursue a PhD in [Program name]at [University name], where I intend to study questions at the intersection of public, labour and health economics. My specific research agenda asks how large-scale government programs can better achieve their desired policy objectives, and the externalities created by such programs in the pursuit of these goals.

So far, I have begun to tackle this research agenda in two key ways: in my undergraduate Honours thesis, my predoctoral position and in independent research. Firstly, in my pre-doctoral research, I am examining how the introduction of Australia's National Disability Insurance Scheme (NDIS) changed the returns to disability care, the composition of the labour market, and potentially the quality of disability care. Secondly, in my independent research, I have been exploring the efficacy of large-scale supply-side education policies on improving individual employment outcomes as well as labour market shortages. Methodologically, I seek to combine causal inference techniques with intuition from theoretical models.

**Academic background.** My academic background demonstrates that I possess the necessary skills to succeed in graduate study. I completed a Bachelor of Commerce (Honours) majoring in Economics and Finance at the University of Melbourne, achieving First Class Honours and receiving numerous awards for my academic performance. My Honours thesis which evaluated the effects of unconventional monetary policy on financial stability in an emerging market context, was also of a high-enough quality to be published as a central bank working paper. I have also achieved excellent grades in proof-based and other mathematics subjects, including Real and Complex Analysis (MATH2023) and Linear Algebra (MATH1002).

**Pre-doctoral and professional experience.** I have been broadening my research toolkit working for Professor Manasi Deshpande as a pre-doctoral researcher. Our project examines the labour supply effects of NDIS, which is a first-of-its-kind disability support program, where the federal government provides a budget to individuals with disabilities based on their needs, which can be spent in the private market on services like personal care, household tasks and physical/behavioural therapy. We exploit the staggered rollout of NDIS across Australia to estimate the causal effects on wages and labour supply. We find that the marginal entrants induced by NDIS into disability work are low-income, lower-skilled women who obtain modest earnings increases from entry. Through this research I have gained significant experience working with large administrative datasets and independently applying an array of causal inference methods such as staggered rollout event studies, DiD and shift-share instruments, as well as using LASSO to define a base population of interest. I have also gained valuable skills in how to carefully implement these econometric methods in ways that ensure credible identification, all of which I believe have strongly prepares me to conduct independent research that appeals to the frontier of the discipline.

Prior to my pre-doctoral position, I worked at the Reserve Bank of Australia, Australia's equivalent of the Federal Reserve. In this role, I led an array of analytical research, most recently running a contingent valuation experiment in a large-scale population representative survey, which is in the process of becoming a central bank working paper.

**Future research.** While auditing Professor Deshpande's course 'Empirical Analysis of Social

Insurance', I became interested in how social transfers target cohorts when need is unobservable (Akerlof, 1978; Nichols and Zeckhauser, 1982), and the challenge in balancing efficient targeting with creating barriers to take-up among intended cohorts (Deshpande and Li, 2019). I was also curious about how I could apply findings from this literature in social insurance to government programs more broadly. One such case study is an Australian education policy called 'Free TAFE', where from 2019 the state of Victoria offered a free vocational education for certain courses that were deemed to have labour shortages.[1] This policy is unique in that eligibility is governed by what courses are available rather than at the recipient level (e.g., by means-testing or employment history).

At first glance, it is unclear which approach is more effective. On the one hand, course-level targeting could achieve two goals simultaneously: like other supply-side policies, it reduces education costs that can deter low-income, low-education individuals with high marginal returns from enrollment, and attempts to maximise the likelihood this education translates to employment by prioritising courses with labour shortages. It is also plausibly easier to implement, given labour shortage sectors is more observable than recipients' underlying need. However, without individual-based targeting, such programs rely on efficient self-sorting which makes it possible that cohorts such as low-skilled, low-education individuals for whom the marginal return to this education is higher could be crowded out of program spots by individuals with low opportunity costs to enrolment, but observably lower need or returns to undertaking this training.

Therefore, in this research, I ask two questions: firstly, is preferentially subsidised training is a legitimate lever for upskilling individuals into growing sectors when implemented at scale? And secondly, does course-based targeting produce similar outcomes when compared to the literature on participant targeting? I leverage variation across states and compare outcomes between Victoria, where free TAFE was introduced, to other states over the years when they had not introduced similar policies. I have been using linked individual administrative data from the Australian Bureau of Statistics, which allow me to observe individuals' prior education history, socio-economic background, parental income, health, and employment outcomes over time.

On my first question, I am interested in whether the composition of the labour market and wages in targeted sectors changed. To do this, I have to find a plausible 'risk set' of individuals who based on ex-ante characteristics are likely enrollees into these courses. In my work thus-far, I find that young women previously on unemployment or family-based income support have the strongest enrolment first-stage.

To answer my second question on targeting, I compare whether the program changed the composition of enrolments in these courses across age, gender, labour force status at enrolment, and highest education level. So far I have found that free TAFE induced higher enrolments among women and working-age cohorts and induced higher enrolments in longer-term courses such as Certificate IV courses and diplomas relative to entry level courses that are likely lower cost. Inspired by the targeting efficiency work in Deshpande and Li (2019), I intend to explore how well this course-based targeting measure fared in inducing likely 'target cohorts' to enter the program (e.g., low-skilled, low-income cohorts), dissuaded individuals with low marginal returns to this education from enrolling, and how much of the enrolment effect comes from inframarginal enrolees.

[1] Examples include disability care, nursing, construction, and early childhood care. From 2023 onwards, this program was phased in by other states.